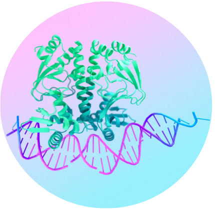
SciLifeLab

# AI agent collaboration with the Model Context Protocol (MCP)

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# What you will build today...

- ❑ A drug discovery **MCP server** exposing tools
- ❑ A **MCP client** that discovers and calls tools dynamically
- ❑ **Security**: detect and defend against tool poisoning attacks
- ❑ **Guardrails**: input validation, blocked compounds, rate limiting
- ❑ **Integration**: connect your earlier **LangGraph agent** to MCP
- ❑ **Bonus 🎁: MCP Python SDK,** from basic to production
- ❑ **Bonus 🎁: integrating with real-world research works!**

# Outline: The bigger picture

## SESSION 2 — BUILDING THE MCP PIPELINE

**1**

### Raw Protocol

Understand MCP from scratch: JSON-RPC, resources, tools

Parts 1 – 2

READING

**2**

### Working Server

Build your own drug discovery MCP server & client

Part 3

HANDS-ON

**3**

### Safe Server

Add guardrails, security layer, and tool poisoning defence

Parts 4 – 5

HANDS-ON

**4**

### Connected Agent

LangGraph + MCP working together end-to-end

Part 6

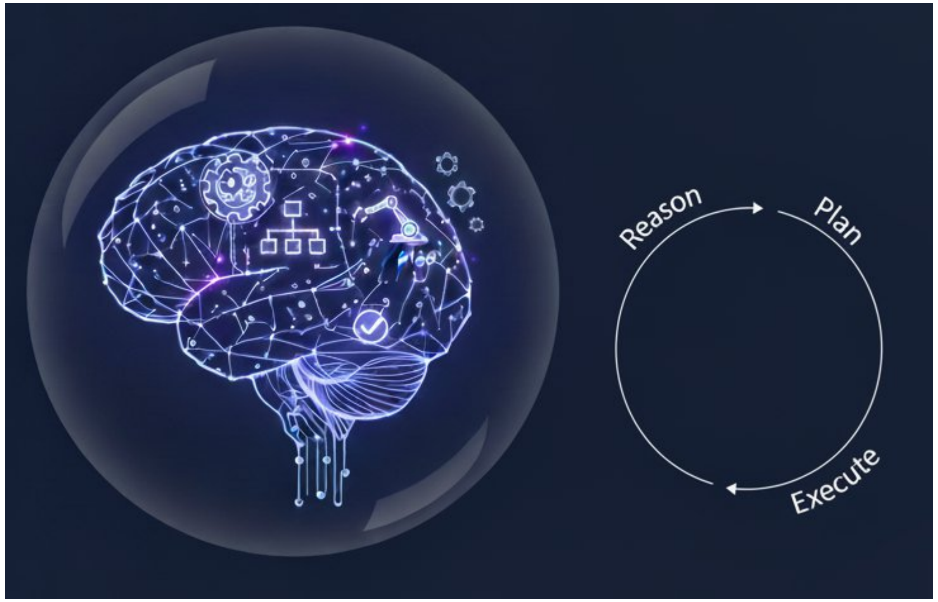
HANDS-ON

*By the end of this session, you will have built every stage of this diagram.*

[1]

# Our journey so far (session 1 recap)

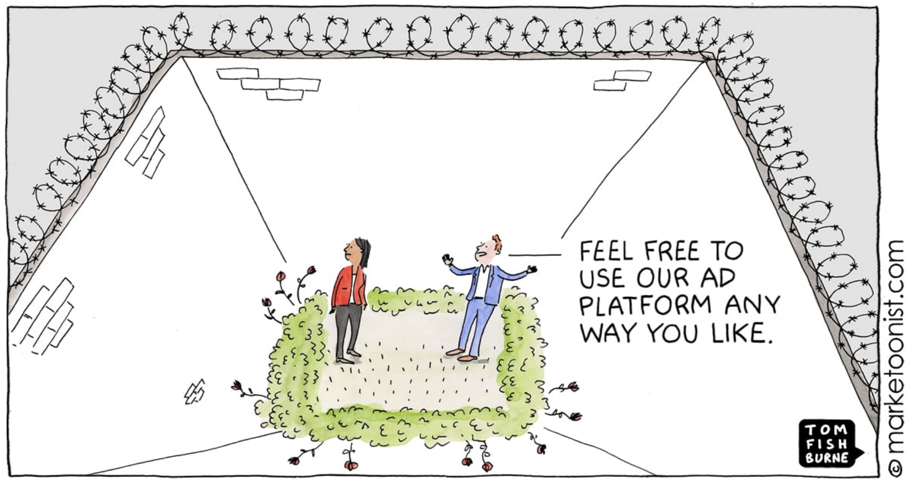
- In Session 1 you built a **LangGraph** drug discovery agent
- Three tools: **`resolve_smiles`**, **`get_properties`**, **`lit_search`**
- It can **reason, plan, and execute** a sequence of actions
- A working proof-of-concept for agentic AI in life sciences




[2]

# But it was a walled garden!

- Your tools were **hardcoded** inside your Python file
- `resolve_smiles`, `get_properties`, and `lit_search` **only worked in your notebook**
- Any other agent or researcher had to **copy** your code **to use** them
- Perfect for a demo, but not for a real research environment

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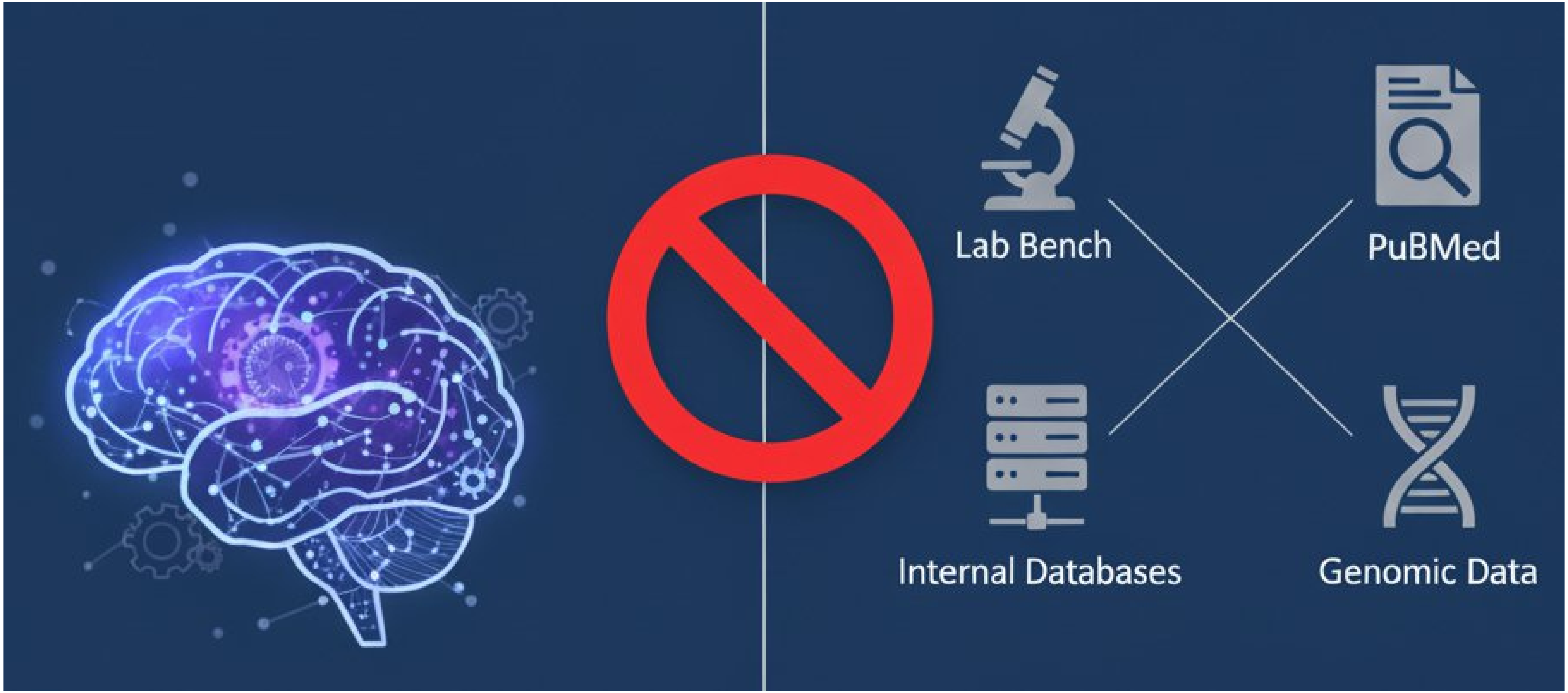
# The Real-world challenges: scalability & reinventing the wheel



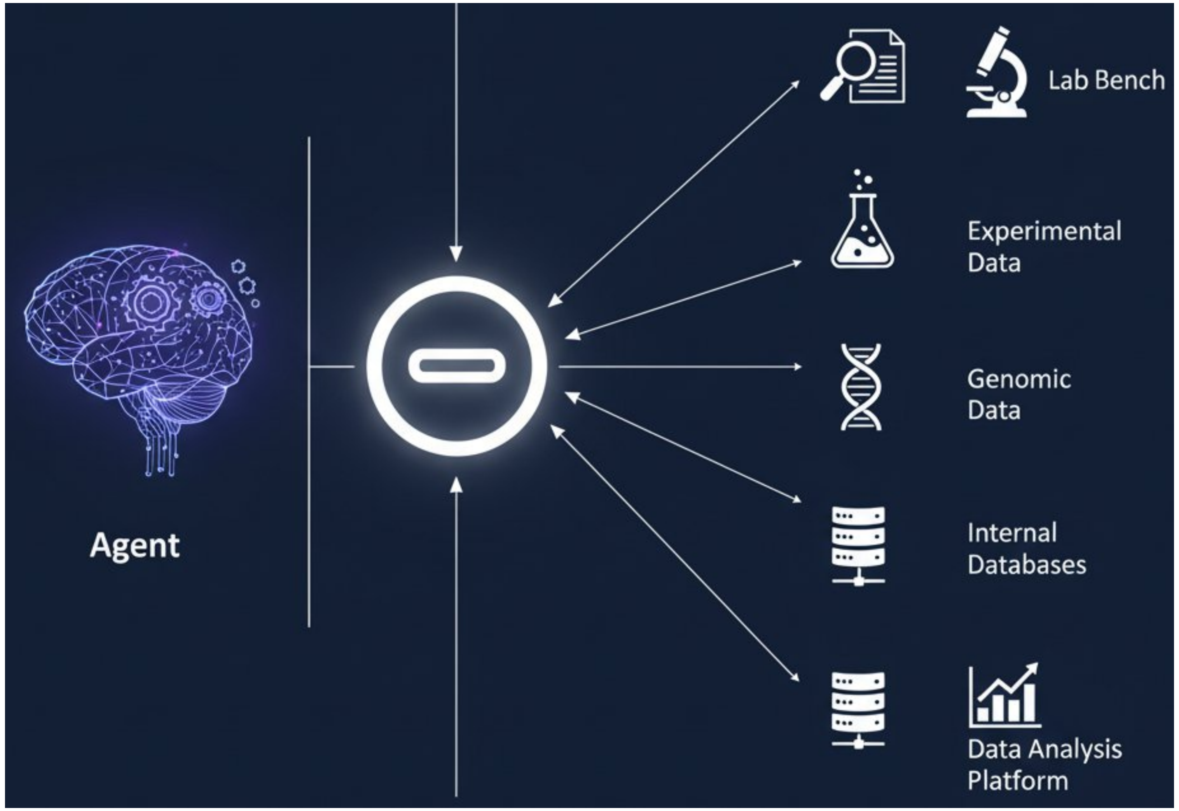

[2]

# The Real-world challenges: scalability & reinventing the wheel

- Need a **new tool** (e.g. ChEMBL, a private drug database)? → **Stop, code, redeploy**
- **Not flexible enough** for fast-paced drug discovery
- Every team builds **isolated agents** and **duplicates** effort
- Teams build their own agents and tools **independently**
- **Duplication, inconsistency, maintenance** burden
- **No standard way to share** or discover tools across groups

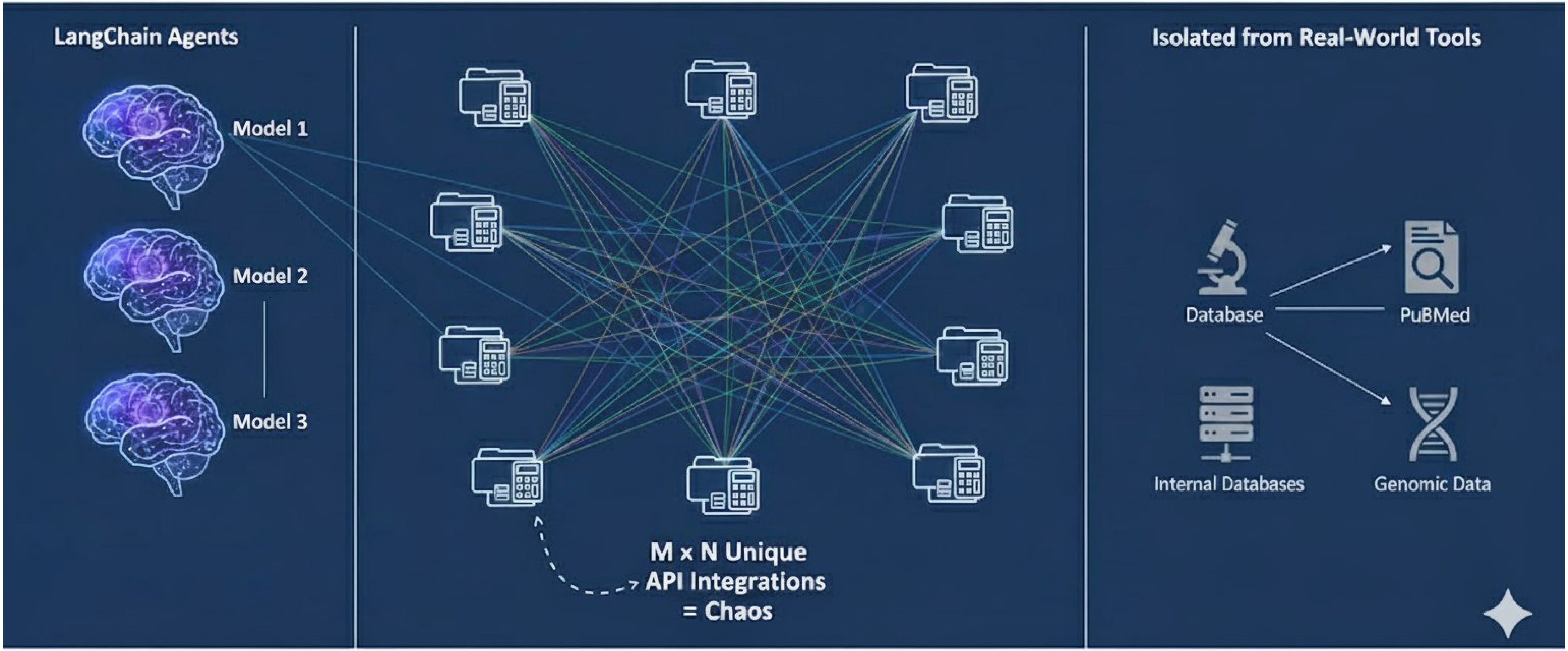
# Solution: introducing MCP (Model Context Protocol)!

- LangGraph = **The brain** (reasoning and planning)
- MCP = **The nervous system** (standardised connections)
- **Dynamic tool discovery**: agents find and use tools automatically
- **Unified ecosystem**: teams build and share tools composably




MCP: The **“USB-C for AI”** [2]

# The M×N integration problem

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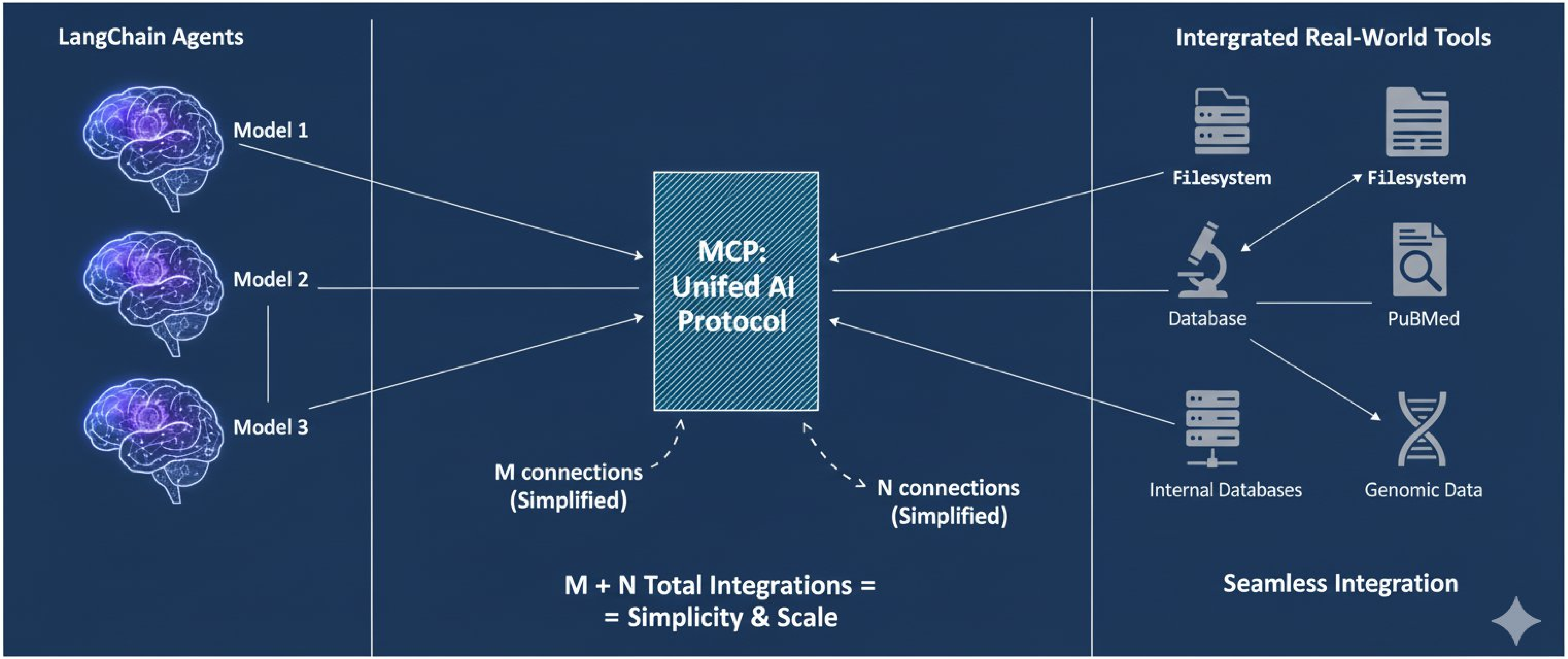
# With MCP (M+N solution)

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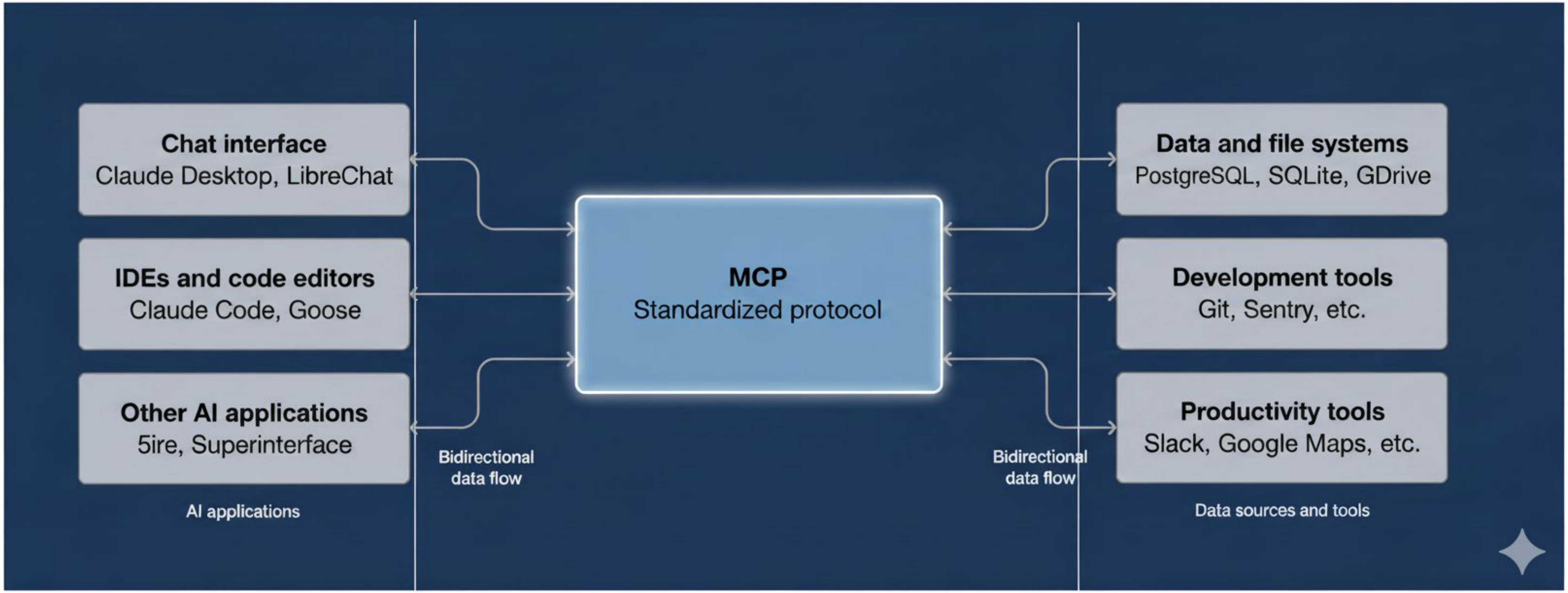
# MCP: overview

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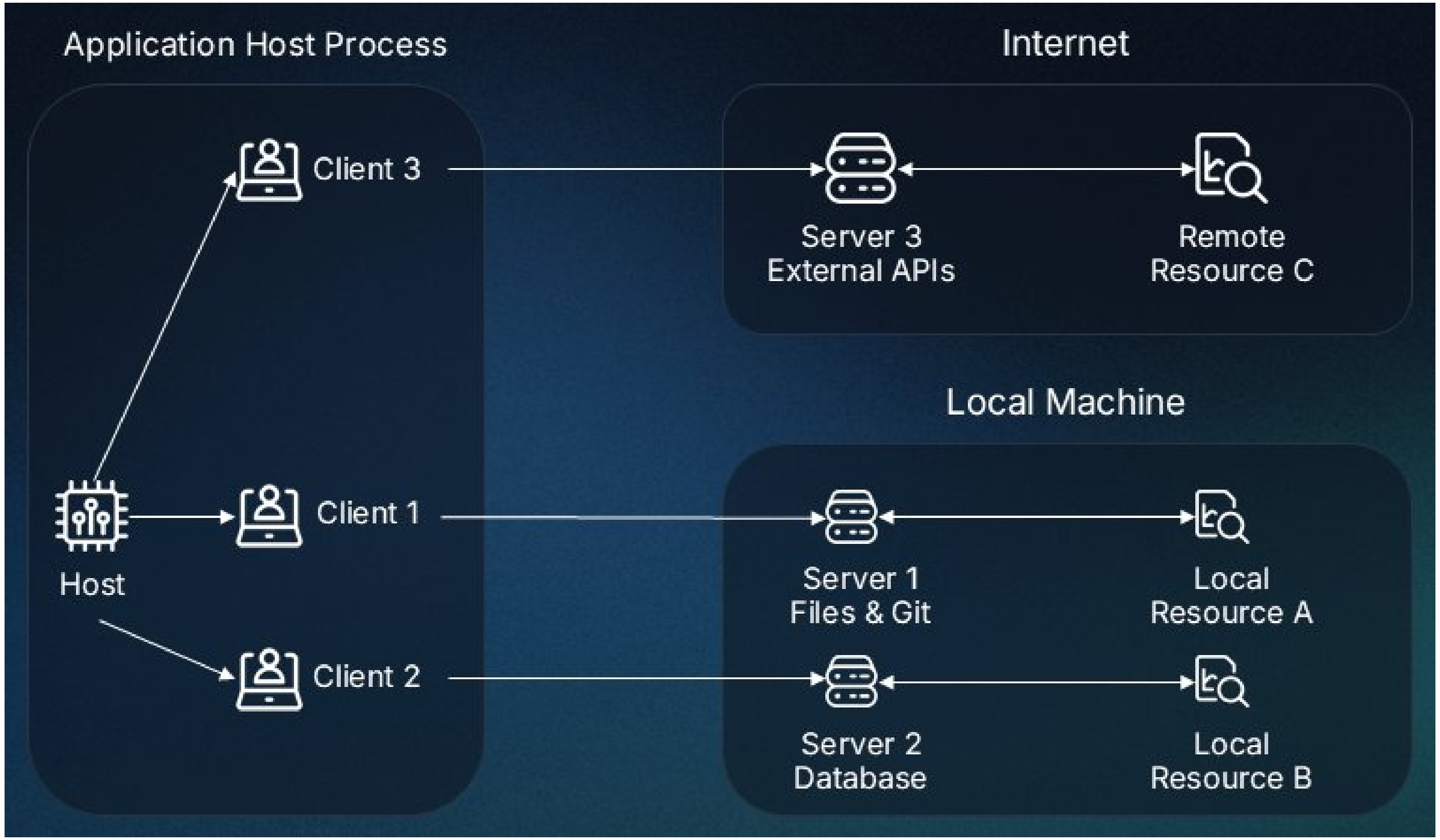
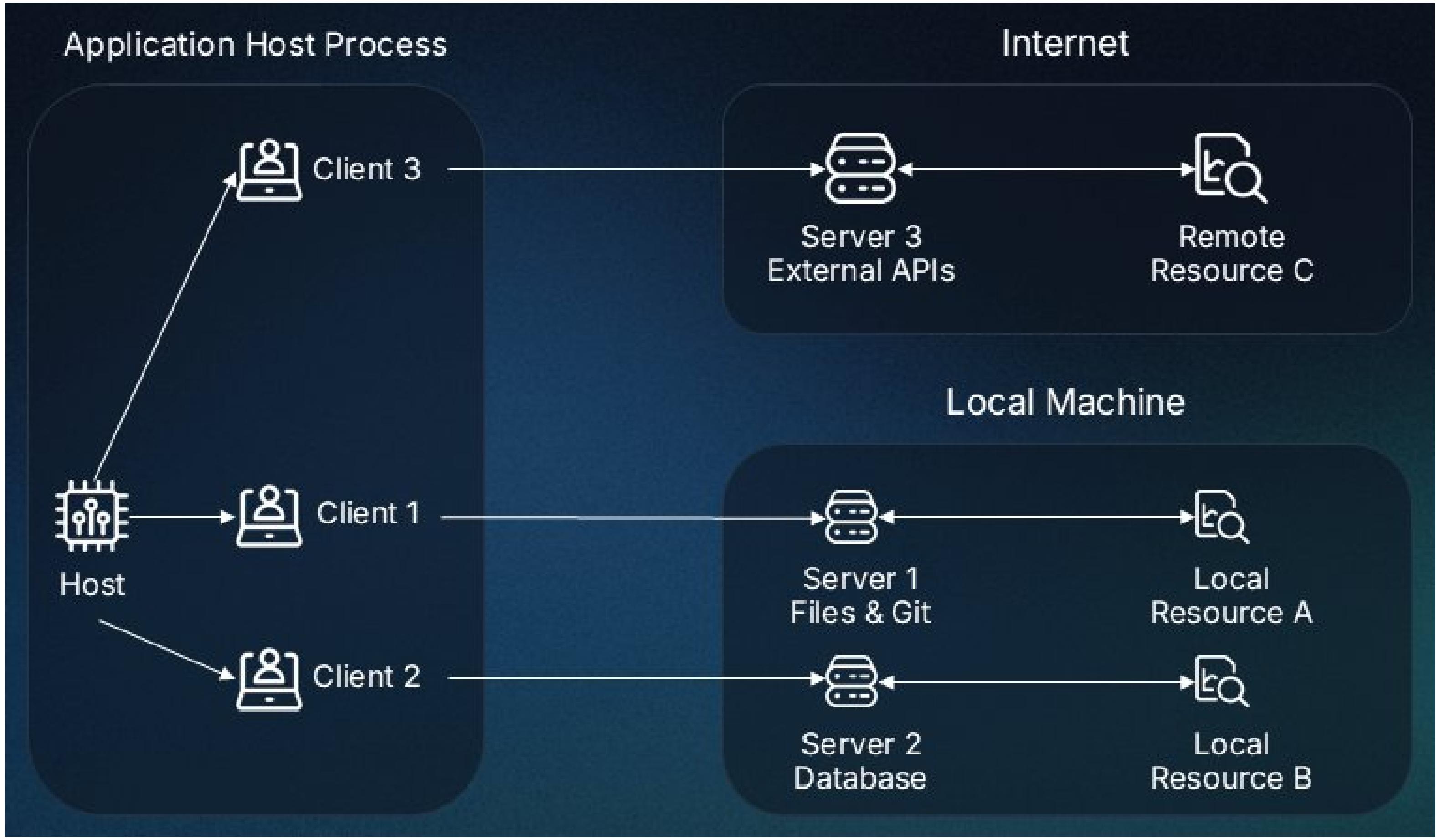
# MCP components: Host




[6]

**Host**: User-facing AI app that interacts with end-users and orchestrates requests, LLM processing, and external tools.

# MCP components: Client

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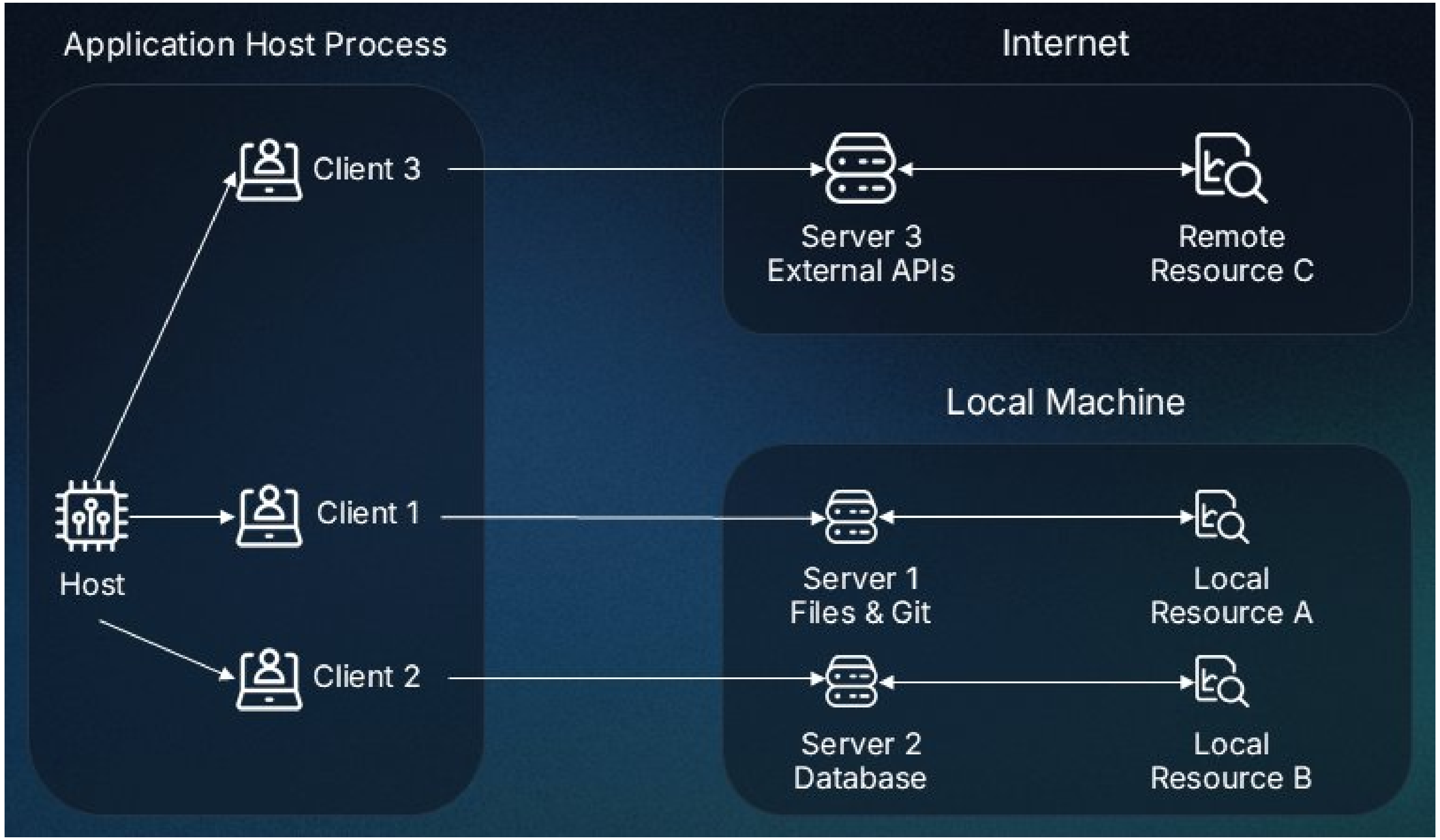
**Client**: Manages communication between the Host and a single MCP Server, handling protocol details.

# MCP components: Server

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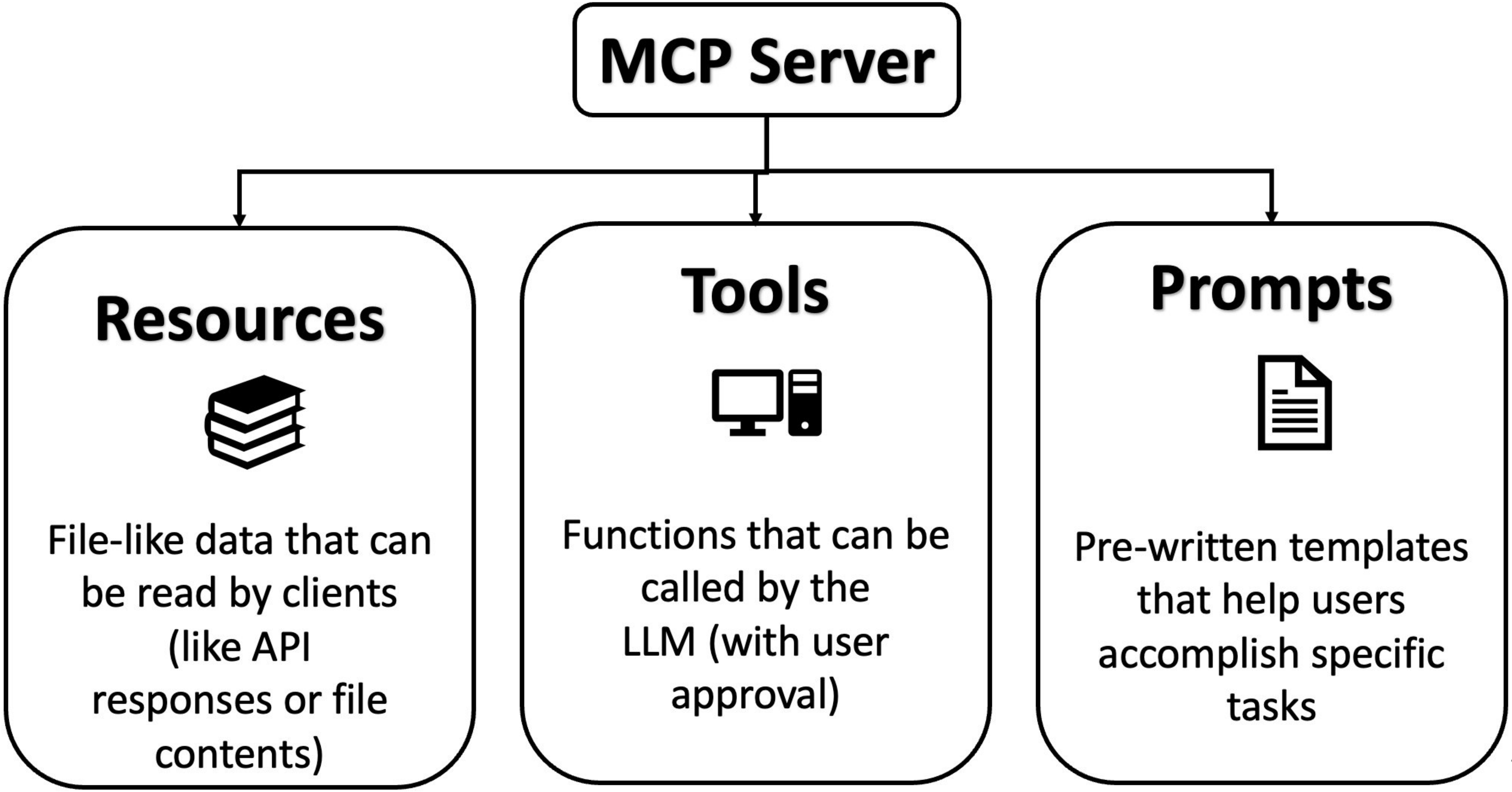
**Server**: External service exposing tools, resources, or prompts via the MCP protocol.

# MCP capabilities: basic

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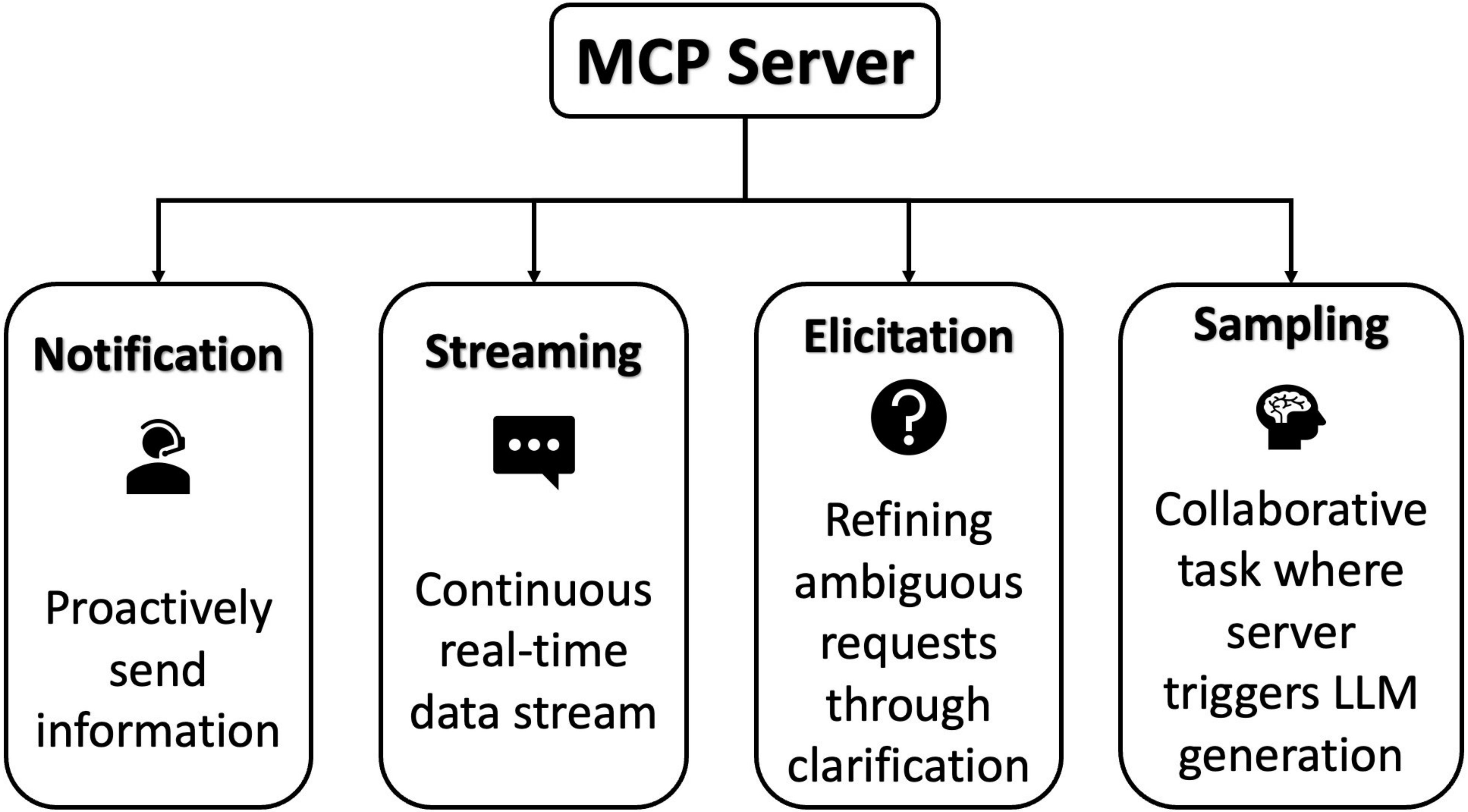
# MCP capabilities: advanced

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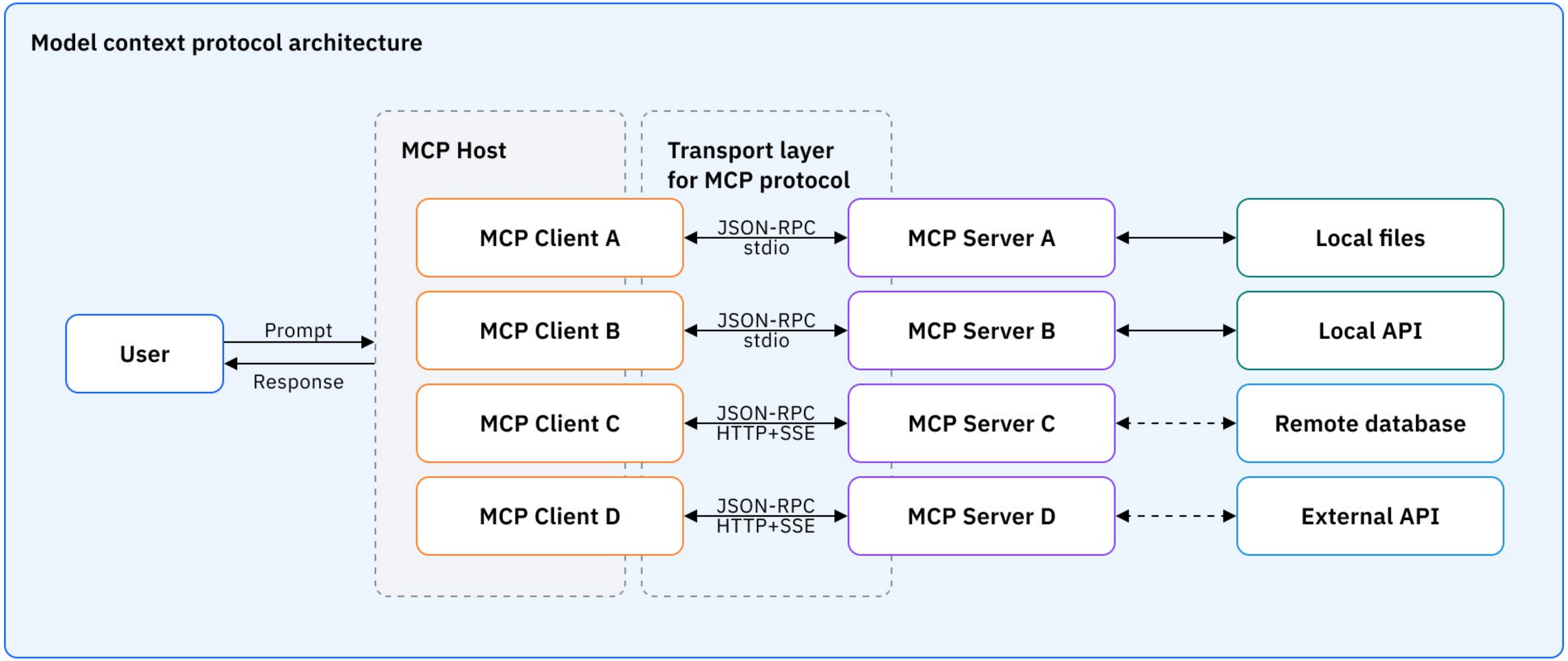
# MCP layers: data layer

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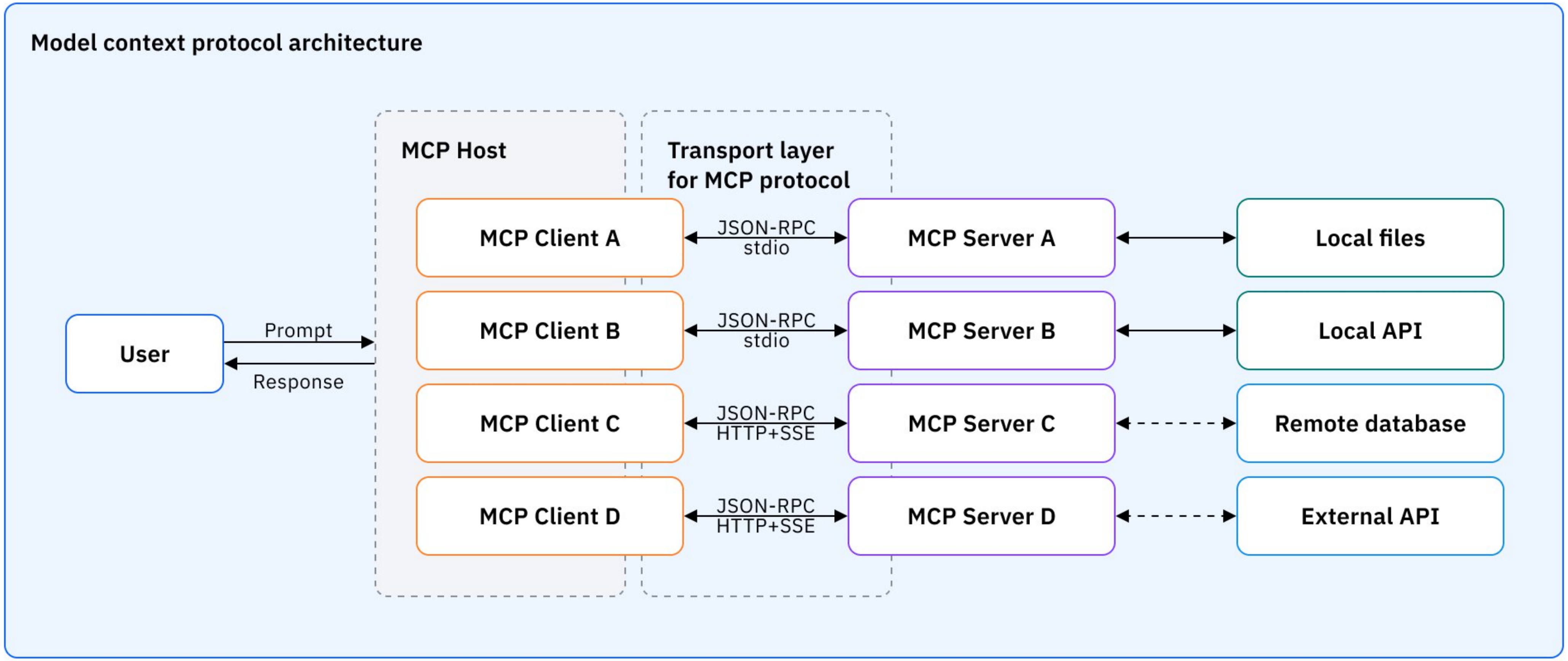
**Data layer:** JSON-RPC protocol defining lifecycle management, tools, resources, prompts, and notifications.

# MCP layers: transport layer

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**Transport layer:** Communication channels, connection handling, message framing, and authentication.

# Why security matters in MCP

- Tool descriptions are read by the LLM and **directly influence** its decisions
- In an open MCP ecosystem you may connect to servers **you didn't write**
- A crafted description is indistinguishable from a legitimate one at the protocol level, this is called **tool poisoning**

# MCP Security Hands On: Tool poisoning & Guardrails

- In **Part 5.3** you will build this **tool poisoning** attack and then defend against it

- **Guardrails** you will implement:
  - ✓ SMILES validation
  - ✓ blocked compounds
  - ✓ rate limiting
  - ✓ description sanitisation

# Bonus 🎁 MCP Python SDK: from basic to production

**Zero boilerplate:** decorators replace manual JSON-RPC handling

**One-line server startup:** mcp.run() vs. custom HTTP/SSE plumbing

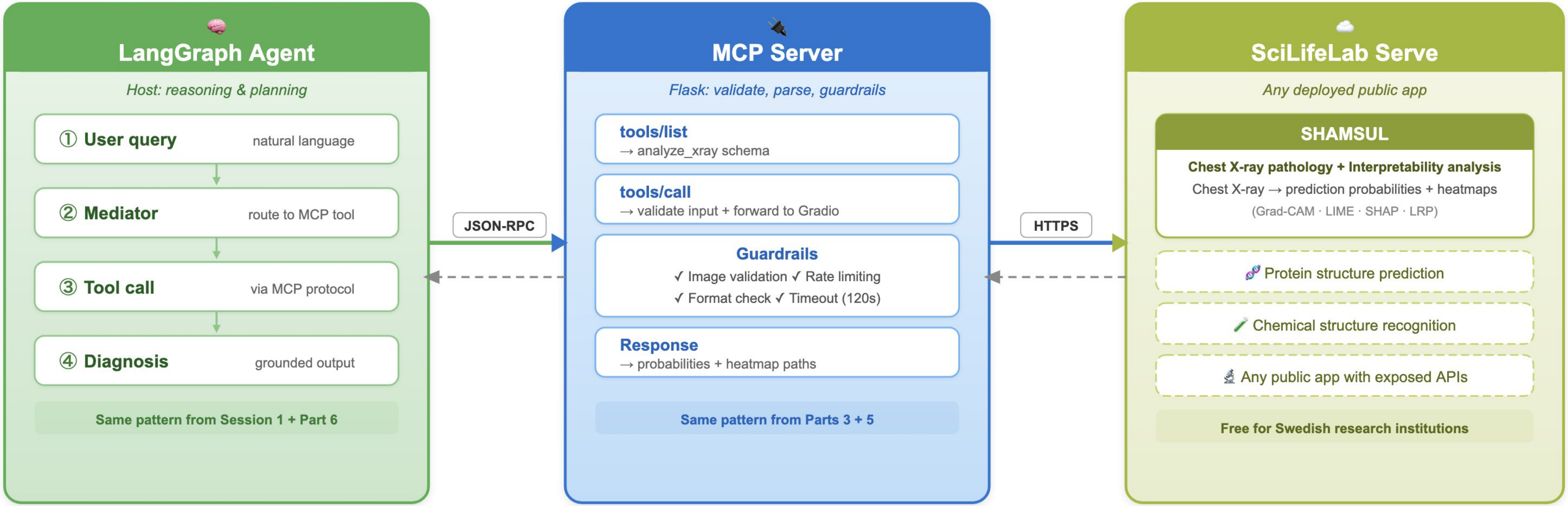
**Built-in LangChain bridge:** no adapter code required

**Type-safe by default:** Pydantic validation out of the box

**SDK speed:** same drug discovery agent, faster implementation

# Bonus 🎁 : SciLifeLab Serve Public App + MCP !

FROM WORKSHOP TO REAL WORLD




**Same MCP pattern you built today — works with any app on SciLifeLab Serve**

serve.scilifelab.se/apps · Bonus_Gradio_MCP/ in the workshop repo

[1]

# Next steps

**shorturl.at/rmCjV**

**Scan or use this link!**

- Make sure your **API key is ready**
- Start and explore! 🏝

# Additional resources

**Official MCP documentation**

https://modelcontextprotocol.io/docs/getting-started/intro

**A nice blog post**

https://www.ramotion.com/blog/mcp-protocol/

**Hugging Face course**

https://huggingface.co/learn/mcp-course/en/unit1/introduction

# References


[1] - AI-generated by Mahbub using Claude Code, Opus 4.6 Extended

[2] - AI-generated by Mahbub using Google Gemini 2.5 Pro

[3] - https://marketoonist.com/2022/05/walled-gardens.html

[4] - AI-generated by Mahbub using Google Gemini 2.5 Pro.
Original idea adapted from https://huggingface.co/learn/mcp-course/en/unit1/key-concepts

[5] - https://modelcontextprotocol.io/docs/getting-started/intro

[6] - https://medium.com/@sathishsuraj/understanding-the-model-context-protocol-mcp-architecture-implementation-and-use-cases-499f1e72673b

[7] - Original idea adapted from https://www.inferless.com/learn/how-to-connect-everyday-tools-with-mcp

[8] - https://www.ibm.com/think/podcasts/mixture-of-experts/ai-action-plan-chatgpt-agents-deepmind-imo

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LangGraph + MCP working together end-to-end

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*You have just built every stage of this diagram. The walled garden is open.*

[1]